Two Measurements per Technician

AP Statistics · Topic 7.4 (CED 4.4) · B: 2027-01-19 · E: 2027-03-05 · TEACHER KEY

CED TETHER

- **Skill 1.E** — Identify an appropriate inference method for significance tests.
- **Skill 1.F** — Identify null and alternative hypotheses.
- **Skill 4.C** — Verify that inference procedures apply in a given situation.
- **EU VAR-7** — The t-distribution may be used to model variation.
- **LO VAR-7.B** — Identify an appropriate testing method for a population mean with unknown σ, including the mean difference between values in matched pairs. [Skill 1.E]
- **EK VAR-7.B.1** — The appropriate test for a population mean with unknown σ is a one-sample t-test for a population mean.
- **EK VAR-7.B.2** — Matched pairs can be thought of as one sample of pairs. Once differences between pairs of values are found, inference for significance testing proceeds as for a population mean.
- **LO VAR-7.C** — Identify the null and alternative hypotheses for a population mean with unknown σ, including the mean difference between values in matched pairs. [Skill 1.F]
- **EK VAR-7.C.1** — The null hypothesis for a one-sample t-test for a population mean is H_0: $\mu = \mu_0$, where μ_0 is the hypothesized value. Depending upon the situation, the alternative hypothesis is H_a: $\mu < \mu_0$, or H_a: $\mu > \mu_0$, or H_a: $\mu \neq \mu_0$.
- **EK VAR-7.C.2** — When finding the mean difference, μ_D, between values in a matched pair, it is important to define the order of subtraction.
- **LO VAR-7.D** — Verify the conditions for the test for a population mean, including the mean difference between values in matched pairs. [Skill 4.C]
- **EK VAR-7.D.1** — In order to make statistical inferences when testing a population mean, we must check for independence and that the sampling distribution is approximately normal: a. To check for independence: (i) Data should be collected using a random sample or a randomized experiment. (ii) When sampling without replacement, check that n $\leq$ 10% N.
- b. To check that the sampling distribution of $\bar{x}$ is approximately normal (shape): (i) If the observed distribution is skewed, n should be greater than 30. (ii) If the sample size is less than 30, the distribution of the sample data should be free from strong skewness and outliers.

Essential question: How does the study structure determine the parameter, hypotheses, and conditions for a mean test?

DOK-3 skill: 4.C · **FRQ pattern:** `paired-test-setup-and-consequences`

DOK rationale: Part (c) coordinates repeated measures, subtraction order, tail direction, independent sample size, and shape evidence, then traces the uncertainty and research-question consequences of the rejected setup.

Do Now (first take) → Explore (video + follow-along) → Exit (finish + turn in) **DOK:** 1 → 3 **Minutes:** 45

Teacher does

- Board slide up at the bell. Ask students to identify the independent observational units.
- Begin the 7.4 video follow-along at minute 5.
- Return at minute 33. Require the unit, subtraction order, tail, and shape evidence.

- Collect at the door. Score part (c) only, E/P/I.

Students do

- Commit to one setup and cite either the measurement structure or claim direction.
- Complete the follow-along, finish (a)–(c), revise, and turn in.

Questions to ask

- How many independent technicians are represented, 18 or 36?
- What sign represents an improvement when $d = \text{new} - \text{old}$?
- What research question would a two-sided alternative answer instead?

Adult role
Solo teacher. Connect the design to one sample of differences and trace uncertainty and tail consequences.

[WARN] WATCH FOR

- Treating repeated measurements on the same 18 technicians as two independent samples.
- Using 36 instead of 18 as the independent sample size.
- Checking the old-time and new-time distributions instead of the distribution of within-technician differences.

SCORING PART (c) — E / P / I

Expected elements

- **chooses-paired-setup** (required) — Uses a matched-pairs one-sample test
- **uses-paired-structure** (required) — Treats 18 technicians as 18 independent units
- **uses-order-and-direction** (required) — Connects new minus old to a lower tail
- **checks-difference-conditions** (required) — Uses the SRS, 10 percent check, and difference shape
- **traces-wrong-procedure** (required) — Diagnoses uncertainty and question-direction consequences

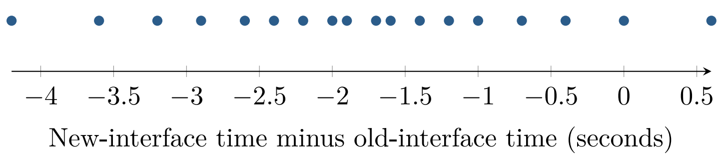
E	Coordinates pairing, subtraction order, lower tail, all conditions, and both consequences of the rejected setup.
P	Recognizes pairing but incompletely explains the tail, conditions, or a consequence.
I	Treats 36 times as independent or chooses a two-sided test without reconciling the reduction claim.

[STAR] TODAY'S PROBLEM — annotated

Suppose a company takes an SRS of 18 of its 300 technicians. Each completes the same task with old and new interfaces; interface order is randomly assigned. The plot shows $d = \text{new time} - \text{old time}$. Researchers ask whether the new interface reduces the population mean time.

Nia: "The technician seems to be the independent unit, so I would build one difference per person; the subtraction order matters for the tail."

Colin: "There are 36 recorded times, so I would use two independent samples of 18 and a two-sided alternative."



FIRST TAKE (5 min)

Which setup better aligns with the study? Cite the measurement structure or the direction of the claim.

Key: Not graded. Students should identify the independent unit and direction before naming a test.

Rules callout "PAIR THE DATA BEFORE YOU TEST (keep on the page)" is printed on the student sheet.

DOK 1 **(a)** Define μ_d and state hypotheses for the research question.

Key: Let μ_d be the population mean new-minus-old time for all 300 technicians. A reduction is negative, so $H_0 : \mu_d = 0$ and $H_a : \mu_d < 0$.

DOK 2 **(b)** Calculate $\bar{d}$ and s_d, verify random, 10 percent, and shape conditions, and name the procedure.

Key: The differences sum to -32.4, so $\bar{d} = -1.8$ seconds; $s_d = 1.2546$ seconds. The SRS satisfies $18 \leq 0.10(300) = 30$. The difference plot is roughly symmetric with no outliers, supporting the small-sample shape check. Use a matched-pairs one-sample t-test on the differences.

DOK 3 **(c)** Adjudicate the setups using the same 18 technicians, subtraction order, directional question, and difference plot. Explain why 36 times are not 36 independent units and how the rejected analysis could change both uncertainty and the question answered.

Key: Nia's setup is warranted. The 18 technicians are the independent units, and their paired measurements form 18 differences. With $d = \text{new} - \text{old}$, reduction means $\mu_d < 0$. The SRS, 10 percent check, and difference plot support the test. Treating 36 linked times as independent discards within-technician pairing and can distort the standard error and p-value. A two-sided alternative asks whether times differ either way, not specifically whether the new interface reduces time.

EXIT

One thing the video changed about my first take: ______________________________